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Project Report
On
Flood management

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Web Technology

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"PROBLEM STATEMENT"

Floods are a natural disaster that can cause immense damage to lives and properties. In recent years, floods have become more frequent and severe due to climate change, urbanization, and other factors. Flood management is a crucial issue that requires proper planning and effective measures to mitigate the impact of floods on people and communities.

However, there is a lack of a centralized platform that provides reliable information on flood management, preparedness, and response. The existing websites are scattered, and the information available is often outdated, incomplete, or not easily accessible to the public. This creates confusion and makes it difficult for people to access the information they need to prepare for floods and stay safe during floods.

Therefore, there is a need for a comprehensive flood management website that provides up-to-date and accurate information on flood risk, preparedness, response, and recovery. Such a website would not only help people stay safe during floods but also assist authorities in making informed decisions about flood management. The website should be user-friendly, accessible, and regularly updated to ensure its effectiveness.

"PROJECT OBJECTIVE"

The objective of the flood management system project is to develop a comprehensive flood management platform that leverages the latest technology and resources to help individuals and communities mitigate the impact of flooding events. The platform will integrate real-time weather data from a current weather API to provide accurate flood warnings and forecasts, and offer a range of features to help individuals and communities prepare for, respond to, and recover from flood events.

To achieve this objective, the project will:

Develop a weather monitoring system that integrates real-time weather data from a current weather API to provide accurate and up-to-date information on weather patterns and potential flood risks. This system will be able to detect potential flood risks early on, allowing individuals and communities to take necessary precautions and preparations.

Establish a news gallery feature on the website that provides up-to-date news and information on flood events, including updates on flood warnings, emergency response efforts, and community impact. This feature will help individuals and communities stay informed and make informed decisions during flooding events.

Create a helpdesk to provide support and assistance to individuals and communities affected by flooding events. The helpdesk will offer guidance on how to prepare for, respond to, and recover from flood events, as well as assistance with accessing financial support and resources. This helpdesk will provide a valuable resource for those affected by flooding events and help foster a sense of community engagement and support.

Establish a donation button on the website to encourage community members to support flood relief efforts and assist those affected by flooding events. This donation button will facilitate

community engagement and encourage support for flood relief efforts, making it easier for individuals to contribute to the cause.

"INTRODUCTION"

Floods are one of the most common and destructive natural disasters worldwide, causing widespread damage to infrastructure, homes, and communities. A flood occurs when water overflows from its natural or artificial boundaries, inundating land that is usually dry. Floods can be caused by a variety of factors, including heavy rainfall, snowmelt, coastal storms, and dam or levee failures. They can happen suddenly and without warning, or they can develop over a period of days or weeks.

Floods can have significant impacts on individuals, communities, and the environment. They can cause property damage, loss of life, and economic disruption, and can have long-lasting effects on the health and wellbeing of affected communities. In addition, floods can damage ecosystems, disrupt agriculture, and contaminate water sources, leading to long-term environmental and public health concerns.

Given the significant impact of floods, it is critical to develop effective strategies and tools for flood management and mitigation. These strategies can include measures such as early warning systems, flood-resistant infrastructure, and community education and preparedness programs. By taking proactive steps to mitigate flood risk and improve flood response, we can help reduce the impact of this devastating natural disaster on our communities and the environment.

FLOOD AND ITS IMPACTS: -

Origin of the word flood stems from the old English word Flood, that is related to the Dutch word vloed and German Flut all of which means "to flow". Flood is the overflow of water beyond a normal level that submerges adjoining land areas which are usually dry. Floods are termed as disasters when submergence in the adjoining areas affects human beings or animals with associated loss. Flood is a hydro-meteorological phenomenon having widespread

occurrence across the globe with varied severity and dimensions. Many of the disaster reports have highlighted the impacts of flood with another high-profile disaster like earthquake and declares that flood incidences in the world are nearly eight times more than that of earthquakes and the number of people affected by floods is nearly 12 times more. There are different perceptions of flood for different stakeholders. For common people, it means devastation, destructions, damage, starvation, loss of lives, damage to properties and infrastructures, etc. Those living in urban areas may treat this phenomenon as disruption in their normal functioning of society. The government machinery and policy planners treat this situation as a factor of retardation in developmental planning with additional overburden on economy and additional expenditure on rescue, relief, rehabilitation, mitigation, etc. The International Commission on Irrigation and Drainage (1995) has defined "flood" as "relatively high flow in a river markedly higher than the usual". The World Meteorological Organization (WMO)/UNESCO International Glossary of Hydrology (WMO 1974) defined flood as "rise, usually brief in the water level of a stream or water body to a peak from which the water level recedes at a slower rate" "relatively high flows as measured by stage height or discharge

TYPES OF FLOODS: -

Floods are classified in various ways depending upon the nature, severity, and sources of inundation. **Riverine flooding** occurs due to various reasons but primarily due to heavy precipitation or glacial melt with resultant runoff. The increased discharge in river channels with decreasing carrying capacities leads to overflow causing inundation in the adjoining low-lying areas. **Flash flooding** is an unprecedented situation that occurs in hilly regions and sloping lands where torrential heavy precipitation, thunderstorm, or cloud burst commonly occurred without any prior warning. This sometimes creates huge loss of lives and damage to properties. **Urban flooding** occurs in regions, where developmental planning has not been in tune with the geo-morphological, ecological, and environmental set up, that results in the increased vulnerability of urban areas. Many urban agglomerations in India are suffering from the problem of flooding even after moderate rainfall. The situation aggravates when rainwater mix up with drain water

causing additional problems including spread of epidemics.

Coastal flooding occurs due to a number of reasons like cyclones and associated storm surge, high tides, tsunami, etc., wherein the low-lying areas in coastal tracts are inundated, as a result of which losses occur on a larger scale. In addition, salinity increases in the coastal groundwater and wells. **Glacial Lake Outburst Flood (GLOF**) occurs in the downstream of glacial regions, where glaciers holding large quantities of water suddenly release them due to melting of ice jam. Glacial outburst is one of the prime reasons of flash floods in some of the Himalayan Rivers. **Cloud Burst Flooding** is the manifestation of climate change and hydrological imbalance that primarily occurs in the form of sudden heavy rainfall. Cyclonic circulations in monsoon may also lead to cloud burst. Cyclone and storm surge flooding mainly occurs in coastal areas due to rainstorms associated with low pressure systems. Movement of cyclonic storms in quick succession leads to severe flooding, especially in low-lying coastal areas.

WHAT AFFECTS FLOODS: -

Floods are affected by many factors which include: Extreme rainfall; Construction of dams; River's behavior and embankments. The other significant reasons include congestion in the drainage channels due to roads, railways, canals, the haphazard growth of industrial and urban establishments may also create flood in a given location. Lack of forests (or deforestation), local water systems, landslides, soil erosion, destruction and/or creation of natural dams, earthquakes, glaciers, sea tides and climate change are some of the factors aggravating of creating flood situation in the country. The flood protection in one area could also result in the floods of other areas. For example, construction of embankments in few areas can increase flood occurrences downstream or upstream. Embankments often act as flood transfer mechanisms as they transfer floods from a given area to their downstream areas. The floods that occur in a given area due to breach of embankments therefore are different to floods in the absence of embankments. During embankment overtopping, the floods are usually sudden, with greater speed, can have increased quantity of sand and could inundate for a longer time as compared floods to during the absence of embankments. Natural dams created by landslides or Glacial Lakes and their outbursts/ overflow

may lead to a large quantity of water flowing in the streams quickly leading to flash floods. In the Himalayan region, such natural dams/ glacial lake overflows during the rainy season could also be aggravated by multiple landslides due to decreased stability of soil.

FLOOD RISK MANAGEMENT: -

Flood disaster refers to the occurrence of a catastrophic flood that brings about adverse impact to humans, animals and the economy. Flood Risk of a particular geographic area or a community refers to the possible consequences of all flood combinations in that particular area. Flood risk management denotes the effort taken by various agencies in managing the risk of floods and the process of reducing the adverse impacts of the flood events (disaster).

In order to have an effective flood risk management in place, there is a need to adopt Flood Risk Management Strategies, that consist of the following components:

FLOOD RISK PREVENTION:-

Aims to decrease consequences of flood event by decreasing exposure of property and people by measures that discourage or prohibit development in flood prone/ high flood risk areas. The focus of this strategy is to “keep people away from water” (Rangier, et al., 2016) by restricting the construction of buildings in flood prone areas.

"TIMELINE OF PROJECT"

Week 1-2

- Project planning and research - Define the project scope, objectives, and requirements. Research current weather APIs and news sources to use in the system.

Week 3-4

- Develop weather monitoring system - Develop the backend system that will retrieve and process real-time weather data from the selected API. Develop the algorithms to analyze and predict potential flooding risks.

Week 5

- Integrate weather API - Integrate the selected weather API into the system and verify the accuracy of the data.

Week 6-7

- Develop news gallery feature - Develop the front-end feature that will display news related to flooding. Gather and organize relevant news sources and design a user-friendly interface for the gallery.

Week 8

- Create helpdesk feature - Develop a helpdesk feature to assist users with any issues or questions related to the system.

Week 9

- Develop donation button - Integrate a donation button into the system to allow users to donate to relevant flood relief organizations.

Week 10-11

Testing and debugging - Test the system thoroughly and identify and fix any bugs or errors.

Week 12

- Deployment and launch - Deploy the system to the web server and launch the system for public use. Promote the system through social media, press releases, and other marketing channels.

"TECHNOLOGY & METHODS USED"

Our flood management system project aims to provide real-time information and resources for individuals and communities affected by flooding. The system utilizes weather APIs to gather real-time weather data and predict potential flood risks. In addition, the system features a news gallery that displays real-time news articles related to flooding. Users can also apply for flood insurance through an integrated form, and make donations through a payment gateway. Helpdesk software is also incorporated to provide users with support and assistance. The system is built using a web development framework, hosted on a cloud-based platform, and managed using an agile project management methodology. Overall, our flood management system aims to provide a comprehensive solution for managing flood risks and assisting those affected by flooding.

Weather APIs - Some popular weather APIs include OpenWeatherMap, Weather Underground, and AccuWeather.

News APIs - Some popular news APIs include NewsAPI, Bing News API, and The New York Times API.

Web development frameworks - Some popular web development frameworks include React, Angular, and Vue.js. You could also use a server-side framework such as Node.js, Django, or Ruby on Rails.

Cloud hosting - Some popular cloud hosting platforms include Amazon Web Services (AWS), Google Cloud Platform (GCP), and Microsoft Azure.

Helpdesk software - Some popular helpdesk software solutions include Zendesk, Freshdesk, and Help Scout.

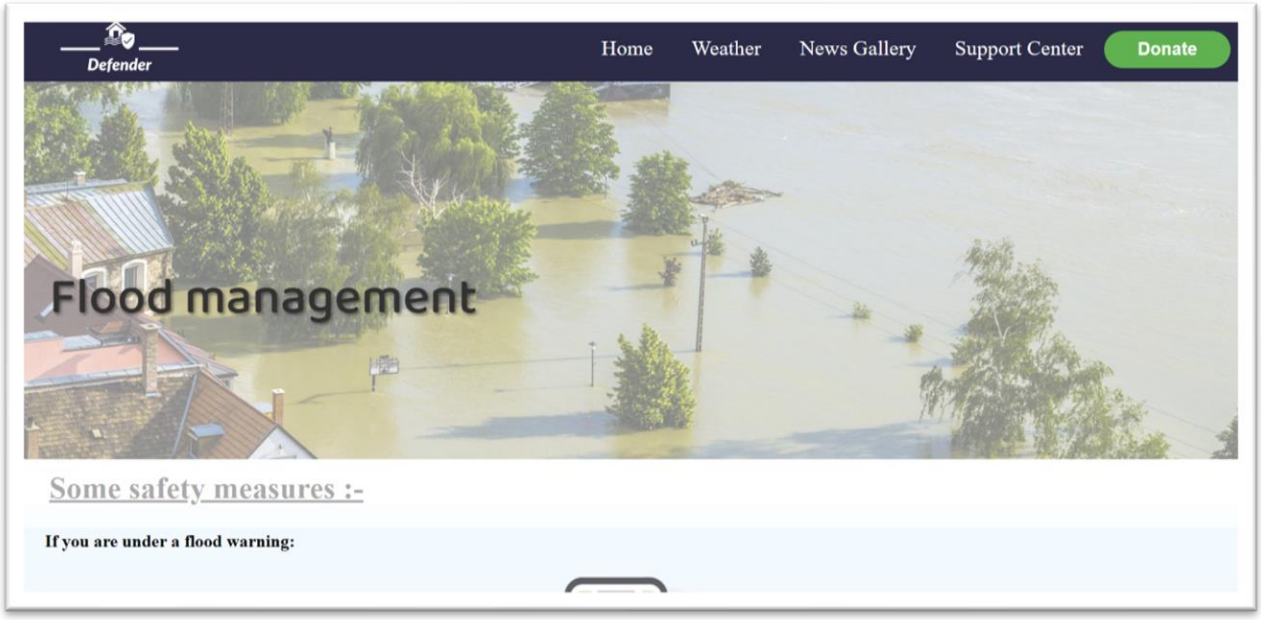
Payment gateway - Some popular payment gateways include PayPal, Stripe, and Braintree.

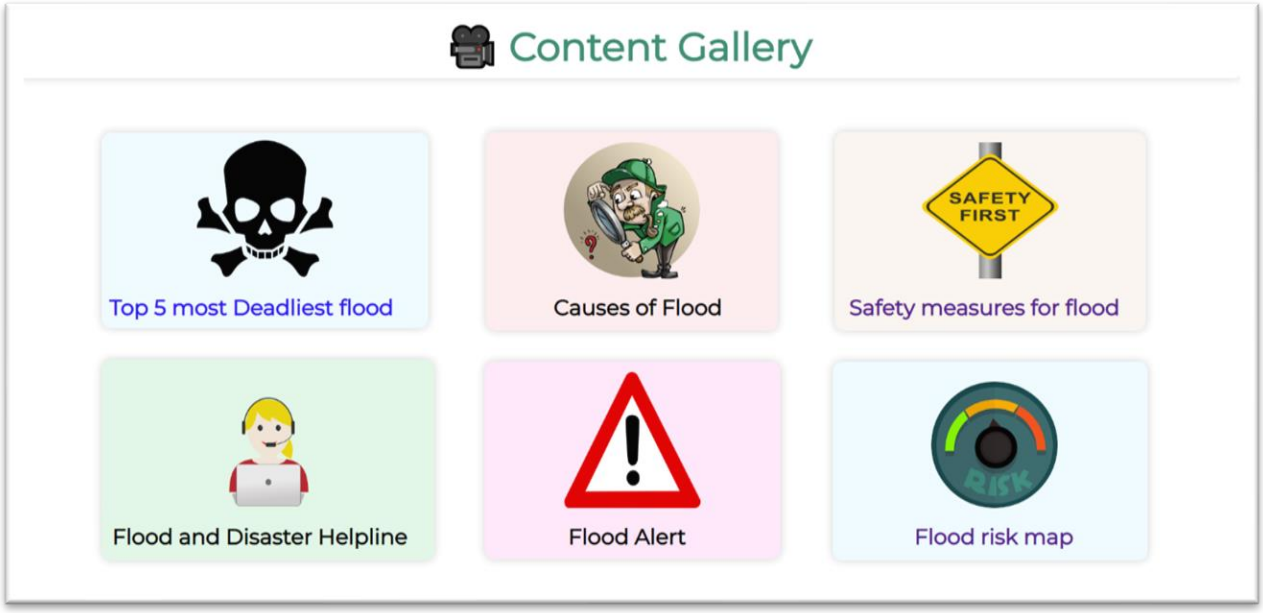
Insurance form - You could use a form-building tool such as Wufoo, JotForm, or Google Forms to build and integrate the insurance form into your system.

Agile project management - Some popular project management tools for agile methodologies include Jira, Trello, and Asana.

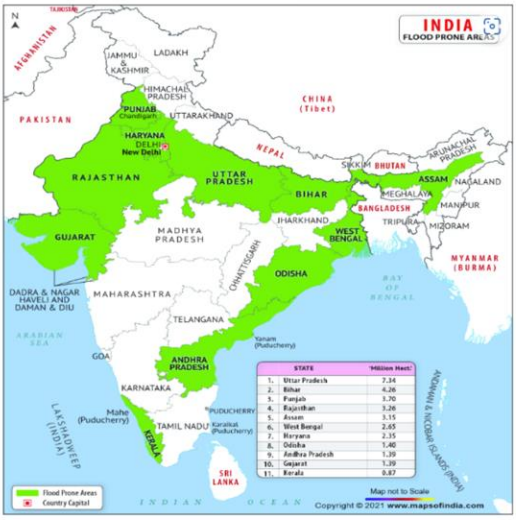
"RESULT"

LANDING PAGE:-





Flood Prones Areas In India



	STATE	Million Hect.
1.	Uttar Pradesh	7.34
2.	Bihar	4.36
3.	Punjab	3.70
4.	Rajasthan	3.26
5.	Assam	3.15
6.	West Bengal	2.65
7.	Haryana	2.35
8.	Odisha	1.40
9.	Andhra Pradesh	1.39
10.	Gujarat	1.39
11.	Kerala	0.87

About floods in India
India Flood Prone Areas
Highest flood prone areas in India

Weather blues continue in California as a storm in the U. state has prompted evacuations. The experts has warn of more floods, while the land is soaked under flood water, a state of emergency has been declared in 43 of California's 58 counties.

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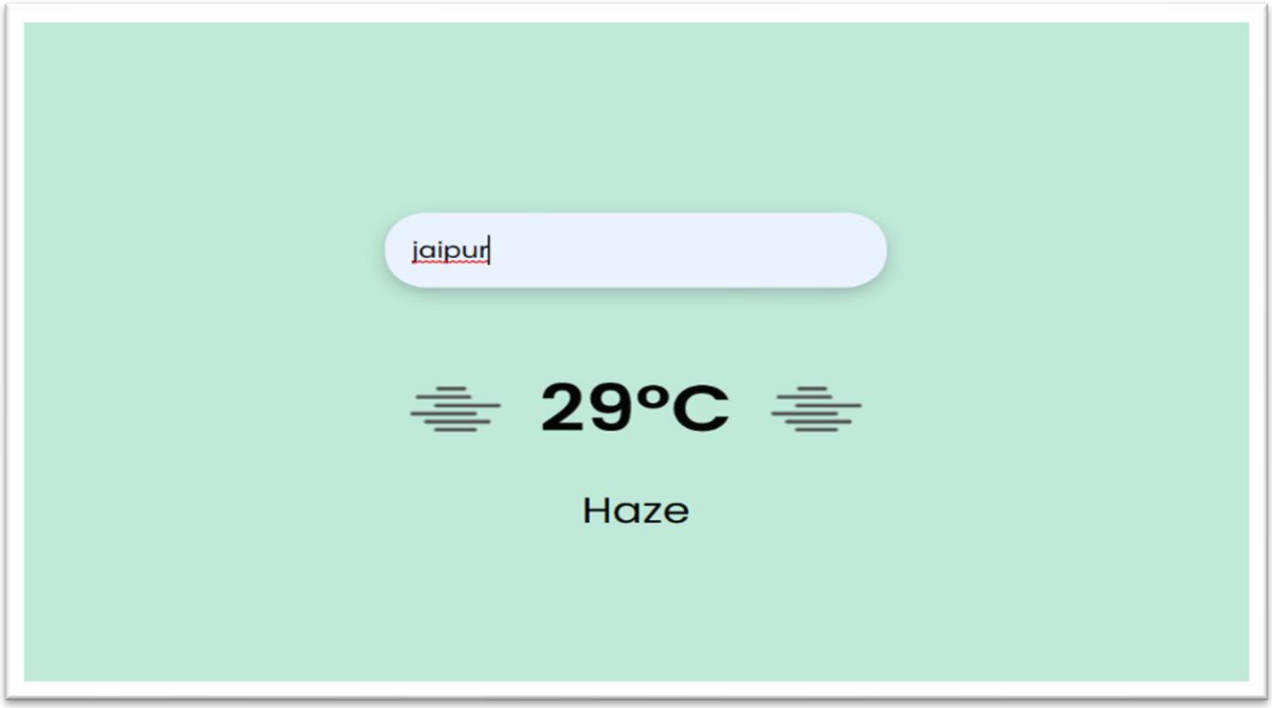
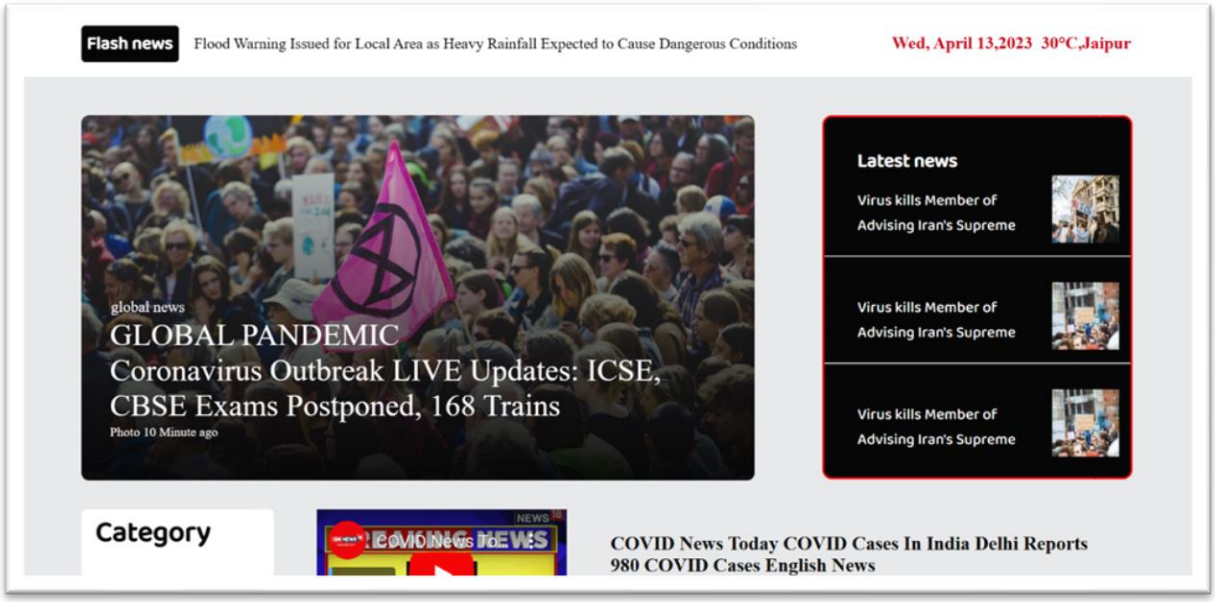
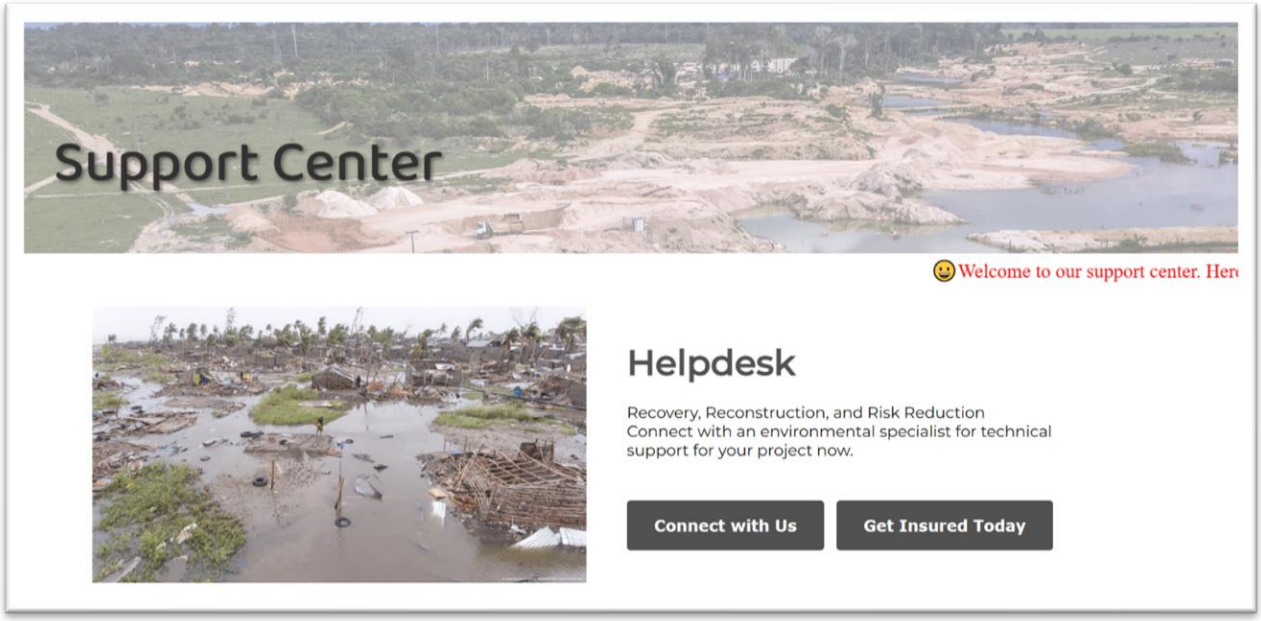
LIVE WEATHER SYSTEM:-



NEWS GALLERY:-



SUPPORT CENTER:-



Life Insurance That Create Wealth

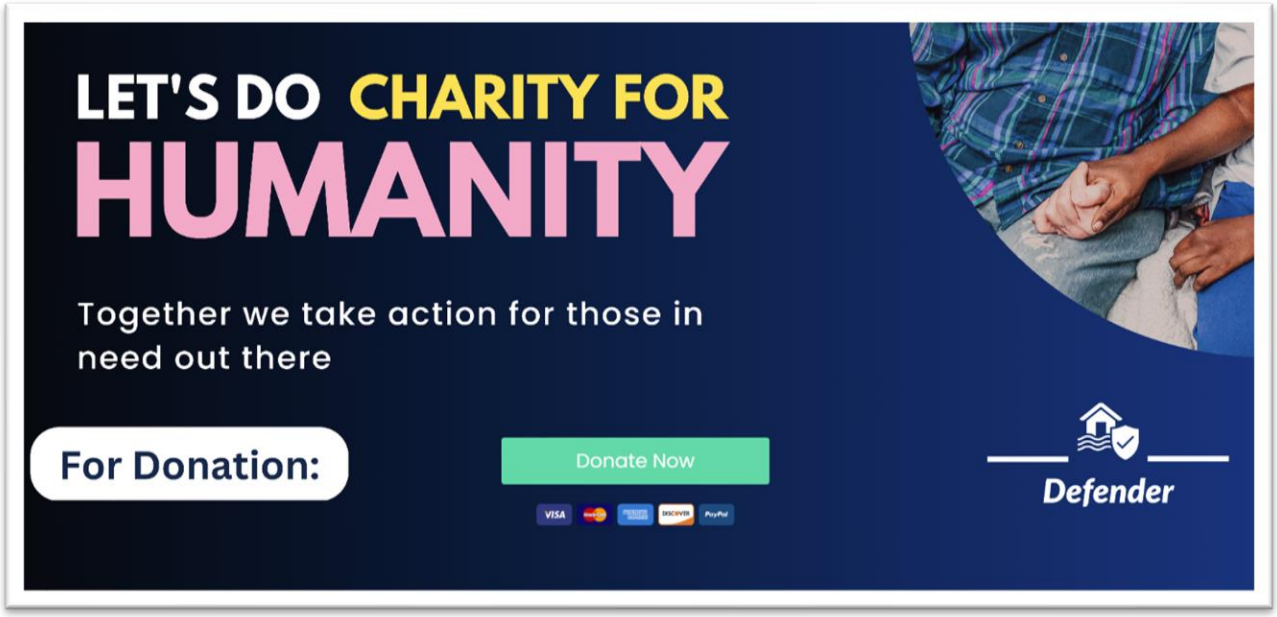
Insurance Details

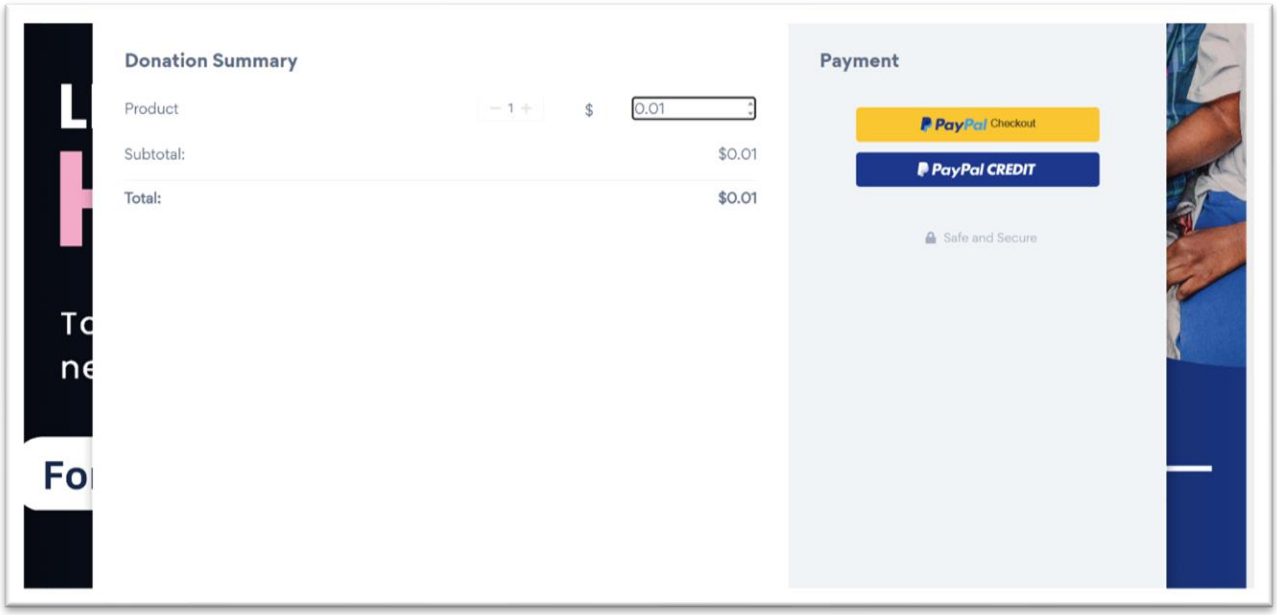
Name:

Email:

Phone:

DONATION:-


LET'S DO CHARITY FOR
HUMANITY
Together we take action for those in need out there
For Donation:
Donate Now
Defender


Donation Summary
Product
1
$
0.01
Subtotal:
$0.01
Total:
$0.01
Payment
PayPal Checkout
PayPal CREDIT
Safe and Secure

"CONCLUSION"

In conclusion, a flood management system website is an essential tool for ensuring the safety and well-being of communities in flood-prone areas. The website provides crucial information about flood risks, preparedness, and response measures, enabling residents to take proactive steps to protect themselves and their properties.

Through this website, residents can access real-time data and alerts, allowing them to make informed decisions and take necessary actions during floods. Additionally, the website can serve as a platform for sharing knowledge and best practices related to flood management, facilitating collaboration and cooperation among stakeholders.

Overall, a flood management system website is an indispensable resource for communities at risk of flooding. By providing timely and accurate information and promoting a culture of preparedness, these websites can help minimize the damage caused by floods and safeguard lives and livelihoods.

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"APPENDIX"

List of Abbreviations and Acronyms:-

FEMA - Federal Emergency Management Agency

NFIP - National Flood Insurance Program

UNDRR - United Nations Office for Disaster Risk Reduction

NOAA - National Oceanic and Atmospheric Administration

IFRC - International Federation of Red Cross and Red Crescent Societies

USGS - United States Geological Survey

GFDRR - Global Facility for Disaster Reduction and Recovery

Additional Figures and Tables:-

Figure 1: Map of flood-prone areas in the region

Table 1: Donations received during the past year for flood relief efforts

Table 2: Frequently asked questions about flood management and prevention

Glossary:-

Floodplain: An area of land adjacent to a river, stream, or other water body that is likely to be flooded during periods of high water flow.

Flash flood: A sudden and rapid flooding of a low-lying area, usually caused by heavy rainfall.

Flood warning: A message issued by authorities to warn residents of an impending flood and provide instructions on how to stay safe.

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